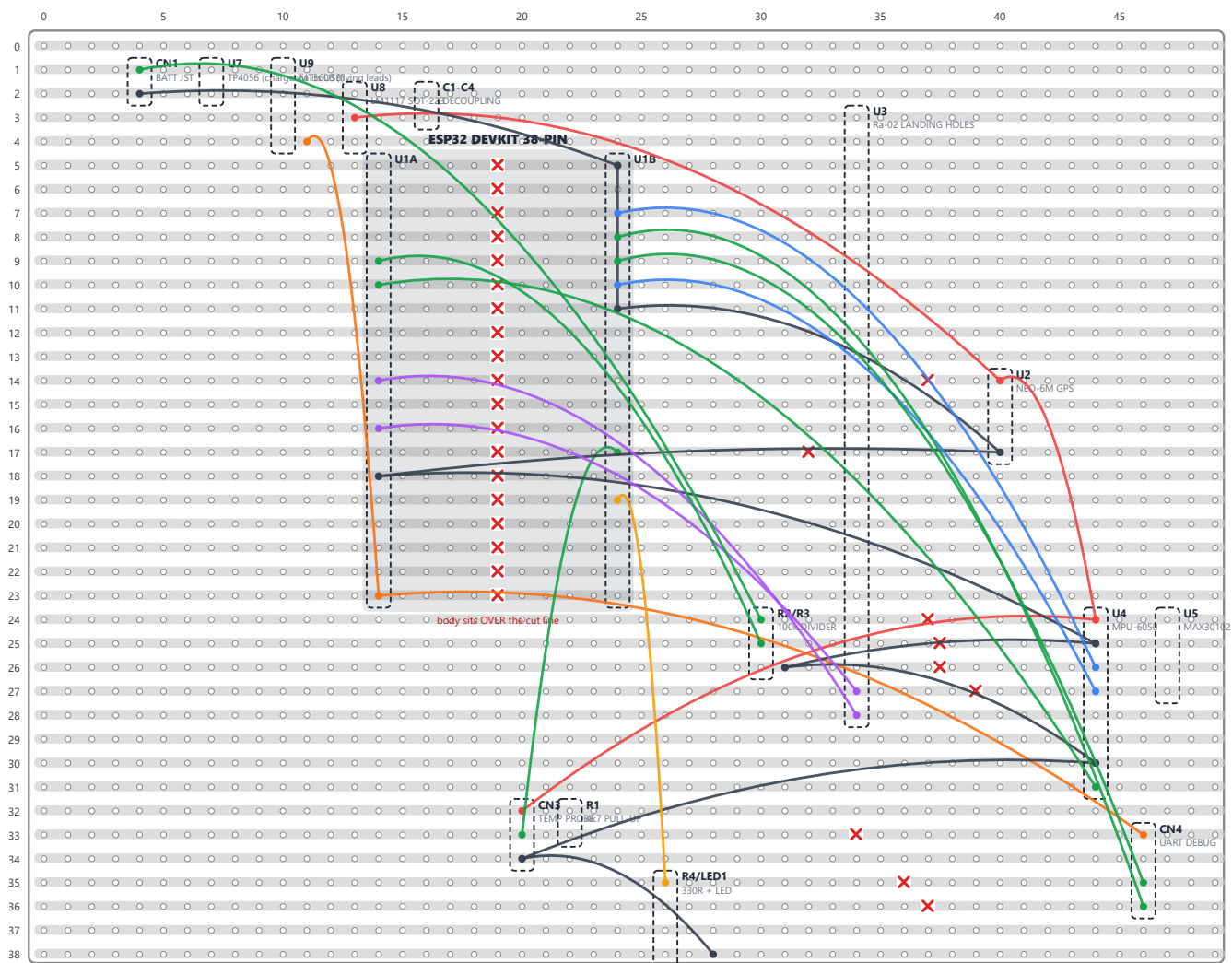


CN-02 Collar Node

Veroboard build book — 28 strip cuts, 25 wire links. Strips run left-right.



CN-02 COLLAR NODE
28 strip cuts (red X) . 25 wire links . strips run left-right

Hole addresses are (col,row); the numbers run along the top and left edge of the diagram.

Step 1 — cut 28 strips

DO THIS BEFORE ANY PART GOES IN. Once the devkit is fitted you cannot reach the cuts underneath it. Meter every cut: probe either side, it must read open circuit.

☐	Row	Cut between	Separates
☐	5	col 14 and col 24	spare U1A.1 GND
☐	6	col 14 and col 24	spare U1A.2 SPI_MOSI
☐	7	col 14 and col 24	spare U1A.3 I2C_SCL
☐	8	col 14 and col 24	spare U1A.4 TX0
☐	9	col 14 and col 24	VBAT_SENSE RX0
☐	10	col 14 and col 24	MPU_INT I2C_SDA
☐	11	col 14 and col 24	spare U1A.7 GND
☐	12	col 14 and col 24	spare U1A.8 SPI_MISO
☐	13	col 14 and col 24	spare U1A.9 SPI_SCK
☐	14	col 14 and col 24	LORA_DIO0 LORA_NSS
☐	14	col 34 and col 40	LORA_NSS +3V3
☐	15	col 14 and col 24	spare U1A.11 GPS_RX
☐	16	col 14 and col 24	LORA_RST GPS_TX
☐	17	col 14 and col 24	spare U1A.13 DS18B20
☐	17	col 24 and col 40	DS18B20 GND
☐	18	col 14 and col 24	GND spare U1B.14
☐	19	col 14 and col 24	spare U1A.15 LED1_A
☐	20	col 14 and col 24	spare U1A.16 spare U1B.16
☐	21	col 14 and col 24	spare U1A.17 spare U1B.17
☐	22	col 14 and col 24	spare U1A.18 spare U1B.18
☐	23	col 14 and col 24	+5V spare U1B.19
☐	24	col 30 and col 44	VBAT +3V3
☐	25	col 31 and col 44	VBAT_SENSE GND
☐	26	col 31 and col 44	GND I2C_SCL
☐	27	col 34 and col 44	LORA_RST I2C_SDA
☐	33	col 22 and col 46	DS18B20 +5V
☐	35	col 26 and col 46	LED1_A RX0
☐	36	col 28 and col 46	K1 TX0

Step 2 — strip map

CHECK THIS BEFORE SOLDERING INTO ANY NEW HOLE. Find the row, find the column range: that is what the hole is already joined to. Rows shaded pink must stay empty.

Row	Cols	Net	Pins in this segment
1	0-49	VBAT	CN1.1@c4 U7.B+@c7 U7.OUT+@c8 U9.IN+@c10
2	0-49	GND	CN1.2@c4 U7.B-@c7 U7.OUT-@c8 U9.IN-@c10 U9.OUT-@c11 U8.1@c13 C1.1@c16 C2.1@c17 C3.1@c18 C4.1@c19
3	0-49	+3V3	U8.2@c13 C1.2@c16 C2.2@c17 C3.2@c18 C4.2@c19 U3.VDD@c34
4	0-49	+5V	U9.OUT+@c11 U8.3@c13
5	0-19	LEAVE EMPTY – devkit 3V3	U1A.1@c14
5	20-49	GND	U1B.1@c24
6	0-19	LEAVE EMPTY – devkit EN	U1A.2@c14
6	20-49	SPI_MOSI	U1B.2@c24 U3.MOSI@c34
7	0-19	LEAVE EMPTY – devkit VP/I036	U1A.3@c14
7	20-49	I2C_SCL	U1B.3@c24
8	0-19	LEAVE EMPTY – devkit VN/I039	U1A.4@c14
8	20-49	TX0	U1B.4@c24
9	0-19	VBAT_SENSE	U1A.5@c14
9	20-49	RX0	U1B.5@c24
10	0-19	MPU_INT	U1A.6@c14
10	20-49	I2C_SDA	U1B.6@c24
11	0-19	LEAVE EMPTY – devkit I032	U1A.7@c14
11	20-49	GND	U1B.7@c24
12	0-19	LEAVE EMPTY – devkit I033	U1A.8@c14
12	20-49	SPI_MISO	U1B.8@c24 U3.MISO@c34
13	0-19	LEAVE EMPTY – devkit I025	U1A.9@c14
13	20-49	SPI_SCK	U1B.9@c24 U3.SCK@c34
14	0-19	LORA_DIO0	U1A.10@c14
14	20-37	LORA_NSS	U1B.10@c24 U3.NSS@c34
14	38-49	+3V3	U2.1@c40
15	0-19	LEAVE EMPTY – devkit I027	U1A.11@c14
15	20-49	GPS_RX	U1B.11@c24 U2.2@c40
16	0-19	LORA_RST	U1A.12@c14
16	20-49	GPS_TX	U1B.12@c24 U2.3@c40
17	0-19	LEAVE EMPTY – devkit I012	U1A.13@c14
17	20-32	DS18B20	U1B.13@c24
17	33-49	GND	U2.4@c40
18	0-19	GND	U1A.14@c14
18	20-49	LEAVE EMPTY – devkit I00	U1B.14@c24
19	0-19	LEAVE EMPTY – devkit I013	U1A.15@c14
19	20-49	LED1_A	U1B.15@c24
20	0-19	LEAVE EMPTY – devkit SD2	U1A.16@c14
20	20-49	LEAVE EMPTY – devkit I015	U1B.16@c24
21	0-19	LEAVE EMPTY – devkit SD3	U1A.17@c14
21	20-49	LEAVE EMPTY – devkit SD1	U1B.17@c24
22	0-19	LEAVE EMPTY – devkit CMD	U1A.18@c14
22	20-49	LEAVE EMPTY – devkit SD0	U1B.18@c24
23	0-19	+5V	U1A.19@c14
23	20-49	LEAVE EMPTY – devkit CLK	U1B.19@c24
24	0-37	VBAT	R2.1@c30
24	38-49	+3V3	U4.1@c44 U5.1@c47

25	0-37	VBAT_SENSE	R2.2@c30 R3.1@c31
25	38-49	GND	U4.2@c44 U5.2@c47
26	0-37	GND	R3.2@c31
26	38-49	I2C_SCL	U4.3@c44 U5.3@c47
27	0-39	LORA_RST	U3.RST@c34
27	40-49	I2C_SDA	U4.4@c44 U5.4@c47
28	0-49	LORA_DIO0	U3.DIO0@c34
30	0-49	GND	U4.7@c44
31	0-49	MPU_INT	U4.8@c44
32	0-49	+3V3	CN3.1@c20 R1.1@c22
33	0-34	DS18B20	CN3.2@c20 R1.2@c22
33	35-49	+5V	CN4.1@c46
34	0-49	GND	CN3.3@c20 CN4.2@c46
35	0-36	LED1_A	R4.1@c26
35	37-49	RX0	CN4.3@c46
36	0-37	K1	R4.2@c26 LED1.A@c28
36	38-49	TX0	CN4.4@c46
38	0-49	GND	LED1.K@c28 U3.GND@c34

Step 3 — every pin, where it goes

☐	Part.pin	Label on part	Net	Hole
☐	C1.1	1	GND	col 16, row 2
☐	C1.2	2	+3V3	col 16, row 3
☐	C2.1	1	GND	col 17, row 2
☐	C2.2	2	+3V3	col 17, row 3
☐	C3.1	1	GND	col 18, row 2
☐	C3.2	2	+3V3	col 18, row 3
☐	C4.1	1	GND	col 19, row 2
☐	C4.2	2	+3V3	col 19, row 3
☐	CN1.1	1	VBAT	col 4, row 1
☐	CN1.2	2	GND	col 4, row 2
☐	CN3.1	1 3V3	+3V3	col 20, row 32
☐	CN3.2	2 DATA (yellow)	DS18B20	col 20, row 33
☐	CN3.3	3 GND (black)	GND	col 20, row 34
☐	CN4.1	1 5V	+5V	col 46, row 33
☐	CN4.2	2 GND	GND	col 46, row 34
☐	CN4.3	3 RX0	RX0	col 46, row 35
☐	CN4.4	4 TX0	TX0	col 46, row 36
☐	LED1.A	A	K1	col 28, row 36
☐	LED1.K	K	GND	col 28, row 38
☐	R1.1	1	+3V3	col 22, row 32
☐	R1.2	2	DS18B20	col 22, row 33
☐	R2.1	1	VBAT	col 30, row 24
☐	R2.2	2	VBAT_SENSE	col 30, row 25
☐	R3.1	1	VBAT_SENSE	col 31, row 25
☐	R3.2	2	GND	col 31, row 26
☐	R4.1	1	LED1_A	col 26, row 35
☐	R4.2	2	K1	col 26, row 36
—	U1A.1	3V3	LEAVE FREE	col 14, row 5
—	U1A.2	EN	LEAVE FREE	col 14, row 6
—	U1A.3	VP/I036	LEAVE FREE	col 14, row 7
—	U1A.4	VN/I039	LEAVE FREE	col 14, row 8
☐	U1A.5	I034	VBAT_SENSE	col 14, row 9
☐	U1A.6	I035	MPU_INT	col 14, row 10
—	U1A.7	I032	LEAVE FREE	col 14, row 11
—	U1A.8	I033	LEAVE FREE	col 14, row 12
—	U1A.9	I025	LEAVE FREE	col 14, row 13
☐	U1A.10	I026	LORA_DIO0	col 14, row 14

—	U1A.11	I027	LEAVE FREE	col 14, row 15
☐	U1A.12	I014	LORA_RST	col 14, row 16
—	U1A.13	I012	LEAVE FREE	col 14, row 17
☐	U1A.14	GND	GND	col 14, row 18
—	U1A.15	I013	LEAVE FREE	col 14, row 19
—	U1A.16	SD2	LEAVE FREE	col 14, row 20
—	U1A.17	SD3	LEAVE FREE	col 14, row 21
—	U1A.18	CMD	LEAVE FREE	col 14, row 22
☐	U1A.19	5V	+5V	col 14, row 23
☐	U1B.1	GND	GND	col 24, row 5
☐	U1B.2	I023	SPI_MOSI	col 24, row 6
☐	U1B.3	I022	I2C_SCL	col 24, row 7
☐	U1B.4	TX0	TX0	col 24, row 8
☐	U1B.5	RX0	RX0	col 24, row 9
☐	U1B.6	I021	I2C_SDA	col 24, row 10
☐	U1B.7	GND	GND	col 24, row 11
☐	U1B.8	I019	SPI_MISO	col 24, row 12
☐	U1B.9	I018	SPI_SCK	col 24, row 13
☐	U1B.10	I05	LORA_NSS	col 24, row 14
☐	U1B.11	I017	GPS_RX	col 24, row 15
☐	U1B.12	I016	GPS_TX	col 24, row 16
☐	U1B.13	I04	DS18B20	col 24, row 17
—	U1B.14	I00	LEAVE FREE	col 24, row 18
☐	U1B.15	I02	LED1_A	col 24, row 19
—	U1B.16	I015	LEAVE FREE	col 24, row 20
—	U1B.17	SD1	LEAVE FREE	col 24, row 21
—	U1B.18	SD0	LEAVE FREE	col 24, row 22
—	U1B.19	CLK	LEAVE FREE	col 24, row 23
☐	U2.1	1	+3V3	col 40, row 14
☐	U2.2	2	GPS_RX	col 40, row 15
☐	U2.3	3	GPS_TX	col 40, row 16
☐	U2.4	4	GND	col 40, row 17
☐	U3.VDD	VDD	+3V3	col 34, row 3
☐	U3.MOSI	MOSI	SPI_MOSI	col 34, row 6
☐	U3.MISO	MISO	SPI_MISO	col 34, row 12
☐	U3.SCK	SCK	SPI_SCK	col 34, row 13
☐	U3.NSS	NSS	LORA_NSS	col 34, row 14
☐	U3.RST	RST	LORA_RST	col 34, row 27
☐	U3.DI00	DI00	LORA_DI00	col 34, row 28
☐	U3.GND	GND	GND	col 34, row 38

☐	U4.1	VCC	+3V3	col 44, row 24
☐	U4.2	GND	GND	col 44, row 25
☐	U4.3	SCL	I2C_SCL	col 44, row 26
☐	U4.4	SDA	I2C_SDA	col 44, row 27
☐	U4.7	AD0	GND	col 44, row 30
☐	U4.8	INT	MPU_INT	col 44, row 31
☐	U5.1	VIN	+3V3	col 47, row 24
☐	U5.2	GND	GND	col 47, row 25
☐	U5.3	SCL	I2C_SCL	col 47, row 26
☐	U5.4	SDA	I2C_SDA	col 47, row 27
☐	U7.B+	B+	VBAT	col 7, row 1
☐	U7.OUT+	OUT+	VBAT	col 8, row 1
☐	U7.B-	B-	GND	col 7, row 2
☐	U7.OUT-	OUT-	GND	col 8, row 2
☐	U8.1	1 GND/ADJ	GND	col 13, row 2
☐	U8.2	2 VOUT (+tab)	+3V3	col 13, row 3
☐	U8.3	3 VIN	+5V	col 13, row 4
☐	U9.IN+	IN+	VBAT	col 10, row 1
☐	U9.IN-	IN-	GND	col 10, row 2
☐	U9.OUT-	OUT-	GND	col 11, row 2
☐	U9.OUT+	OUT+	+5V	col 11, row 4

Step 4 — solder 25 wire links

Insulated wire on the component side. Tick as you go, and buzz each one out after fitting.

☐	Net	From	To
☐	+3V3	col 13, row 3	col 40, row 14
☐	+3V3	col 40, row 14	col 44, row 24
☐	+3V3	col 44, row 24	col 20, row 32
☐	+5V	col 11, row 4	col 14, row 23
☐	+5V	col 14, row 23	col 46, row 33
☐	DS18B20	col 24, row 17	col 20, row 33
☐	GND	col 4, row 2	col 24, row 5
☐	GND	col 14, row 18	col 44, row 25
☐	GND	col 20, row 34	col 28, row 38
☐	GND	col 24, row 5	col 24, row 11
☐	GND	col 24, row 11	col 40, row 17
☐	GND	col 31, row 26	col 44, row 30
☐	GND	col 40, row 17	col 14, row 18
☐	GND	col 44, row 25	col 31, row 26
☐	GND	col 44, row 30	col 20, row 34
☐	I2C_SCL	col 24, row 7	col 44, row 26
☐	I2C_SDA	col 24, row 10	col 44, row 27
☐	LED1_A	col 24, row 19	col 26, row 35
☐	LORA_DIO0	col 14, row 14	col 34, row 28
☐	LORA_RST	col 14, row 16	col 34, row 27
☐	MPU_INT	col 14, row 10	col 44, row 31
☐	RX0	col 24, row 9	col 46, row 35
☐	TX0	col 24, row 8	col 46, row 36
☐	VBAT	col 4, row 1	col 30, row 24
☐	VBAT_SENSE	col 14, row 9	col 30, row 25

Before power

☐	Meter check	Expect
☐	GND to +3V3	open
☐	GND to +5V	open
☐	GND to VBAT / +12V	open
☐	every cut, across it	open